

Minkowski's Theorem

The following result plays an important role in theorems in algebraic number theory that involve (Euclidean) bounds.

A symmetric convex subset $C \subset \mathbb{R}^n$ that has volume greater than 2^n contains a non-zero point of $\mathbb{Z}^n$.

A subset $C \subset \mathbb{R}^n$ is said to be *symmetric* if $-C = C$.

A subset $C \subset \mathbb{R}^n$ is said to be *convex* if, whenever $v, w \in C$, we have $tv + (1-t)w \in C$ for $t \in [0, 1]$.

We will also make use of the following corollary:

A symmetric convex compact subset $K \subset \mathbb{R}^n$ that has volume not less than 2^n contains a non-zero point of $\mathbb{Z}^n$.

We make use of any conventional notion of volume of a subset of $\mathbb{R}^n$. For example, we can take the limit

$$\lim_{k \rightarrow \infty} \frac{|2^k C \cap \mathbb{Z}^n|}{2^{kn}}$$

which works for many subsets such as bounded open sets or their closures.

Proof

Let us divide $\mathbb{R}^n$ into “cubes” of the form $w + [-1, 1]^n$ where $w \in 2\mathbb{Z}^n$; these are translates of the “standard” cube $[-1, 1]^n$ by elements of $2\mathbb{Z}^n$. We note that two such cubes do not overlap, and each has volume 2^n .

Next, for each $w \in 2\mathbb{Z}^n$ let $C_w \subset [-1, 1]^n$ be such that

$$w + C_w = C \cap (w + [-1, 1]^n)$$

Since translation does not change the volume, we see that

$$\text{vol}(C) = \sum_w \text{vol}(C_w) > 2^n$$

Since $C_w \subset [-1, 1]^n$ and the sum of the volumes is greater than the volume of this cube, we must have $v, w \in 2\mathbb{Z}^n$ *distinct* such that $C_v \cap C_w$ is non-empty; say t is a point in the overlap. Thus means that $a = v + t$ and $b = w + t$ are in C . Thus, $a - b = v - w$ is a *non-zero* element of $2\mathbb{Z}^n$.

We have $a, b \in C$ such that $a - b \in 2\mathbb{Z}^n \setminus \{0\}$.

Note that $-b$ is in C since b is in C and C is symmetric. Secondly $(a - b)/2$ is the *mid-point* of the segment joining a and $-b$ in $\mathbb{R}^n$; hence, it is in C since C is convex. This means that we have found a point $(a - b)/2$ which lies in C and in $\mathbb{Z}^n \setminus \{0\}$ as required.

Compact Case

For each $m \in \mathbb{N}$, we apply the earlier result to $(1 + 1/m)K$ (which has volume *greater than* 2^n) to obtain $v_m \in K$ such that $w_m = (1 + 1/m)v_m$ lies in $\mathbb{Z}^n \setminus \{0\}$.

Now (v_m) is an infinite sequence in a compact set K , thus it has a convergent subsequence. In other words, there is an increasing sequence $m_1 < m_2 < \dots$ such that $\lim_{r \rightarrow \infty} v_{m_r} = v \in K$. It follows that $\lim_{r \rightarrow \infty} (1 + 1/m_r)v_{m_r} = v$ as well.

Now (w_{m_r}) is a convergent sequence of points in the *discrete* set $\mathbb{Z}^n \setminus \{0\}$. So we must have some r_0 such that for $r \geq r_0$ we have $w_{m_r} = v$ is a *constant* element of $\mathbb{Z}^n \setminus \{0\}$. We thus have v in K and in $\mathbb{Z}^n \setminus \{0\}$ as required.